RT-BE92U Stock Firmware: SSH Security and Health Audit

Version 1.7.0 · 2026-10-14 · ASUS RT-BE92U · Asuswrt 3.0.0.6.102 (stock)

1. Scope

- Target: ASUS RT-BE92U on stock Asuswrt 3.0.0.6.102; the newest ASUS release as of 2026-10-14 is 102_40497, released 2026-10-12 [R6].
- Coverage: security- and health-relevant settings readable over SSH (nvram and runtime state); fixes use the web UI wherever it offers the setting (section 4.3).
- Out of scope: Asuswrt-Merlin features and custom startup scripts (I-02 only probes for a hook).

2. Summary

ID	SEV	CHECK	PASS WHEN / EXPECTED	STATE	PAGE
C-01	CRIT	SSH backdoor indicators	`sshd_port` ≠ 53282; own keys only		3
C-02	CRIT	SSH reachable from WAN	`sshd_enable` = 0 or 2		4
C-03	CRIT	Web UI reachable from WAN	`misc_http_x` = 0		4
H-01	HIGH	Firmware currency	102_40497 or newer		5
H-02	HIGH	Telnet daemon	`telnetd_enable` = 0		5
H-03	HIGH	WPS	`wps_enable(_x)` = 0; modes off		5
M-01	MED	SSH password login	`sshd_pass` = 0; ≥ 1 key		6
M-02	MED	DoT certificate checks	DoT on, `dnspriv_profile` = 1		6
M-03	MED	DNS rebind protection	`dns_norebind` = 1		7
M-04	MED	Samba guest access	anonymous access refused		7
M-05	MED	UPnP	`wan_upnp_enable` = 0		7
L-01	LOW	Browser auto-DoH bypass	`dns_priv_override` 1 (0 if DoT on)		8
L-02	LOW	Web UI over plain HTTP	`http_enable` = 1		8
L-03	LOW	Admin access allowlist	`enable_acc_restriction` = 1		8
L-04	LOW	Idle timeouts	both non-zero		8
L-05	LOW	DDNS exposure	`ddns_enable_x` = 0		8
L-06	LOW	SSH default port	`sshd_port` 1024–65535, ≠ 53282		9
L-07	LOW	USB SSD TRIM	`discard_max_hw_bytes` > 0; trim ok		9
I-01	INFO	Persistence, reboot schedule	`11111110405`; `/jffs` ro/rw noted		9
I-02	INFO	Boot-hook probe	result recorded		9
I-03	INFO	Thermals, radios, clients	values recorded		10
I-04	INFO	Listening services baseline	baseline saved		10

Severity. CRIT: remote compromise path or active indicator; fix first. HIGH: exploitable exposure or missing security fixes. MED: weakens a control. LOW: hardening or health. INFO: baseline data, recorded but not graded.

State. PASS; FIXED (failed, then fixed and re-verified); ACCEPTED (MED or LOW only: a deliberate exception or a hardware or firmware limit, reason noted under the item); FAIL; RECORDED (INFO rows).

Verdict: PENDING. FAIL if any row is FAIL; PASS-WITH-FIXES if no row is FAIL and at least one is FIXED or ACCEPTED; PASS if every graded row is PASS.

3. Procedure

1. Setup and first snapshot (section 5).
2. Collector (section 6); keep the output.
3. C-01 before any other check. Any indicator: stop, follow its fix, then start again at step 1.
4. C-02 and C-03, then HIGH, MED, LOW and INFO. After each fix, re-run the item's check; a **Verify** line adds any further test.
5. Collector again, then a second snapshot and the diff (section 5).
6. After the next reboot (04:05 or manual), run the collector once more; every value must persist.
7. Fill in the STATE column and the verdict (section 2).

4. Conventions

4.1 Prerequisites

- SSH reachable from the host through the `be92u` host alias (section 5). If SSH is off, turn it on as LAN only; otherwise change no SSH setting before C-01 and the first snapshot record it, even a port of 53282 (a C-01 indicator).
- Where commands run: **router** blocks in BusyBox ash over `ssh be92u`; **host** blocks in fish on the CachyOS host; **GUI** steps in the router web UI at the menu path given.
- Host tools: `ssh`, `ssh-keygen`, `sha256sum`, `diff`, `drill` (ldns) and, for M-04, `smbclient`.

4.2 Placeholders

TOKEN	MEANING	TOKEN	MEANING
`USER`	router admin login name	`FIRMWARE_FILE`	downloaded firmware file
`ROUTER_IP`	router LAN address	`SHARE`	a share name from the M-04 router check
`SSH_PORT`	current SSH port (Administration → System)	`IFACE`	radio interface printed by I-03
`YOUR_KEY`	your SSH private key (`.pub`: public key)	`CLIENT_MAC`	client MAC address from `wl assoclist`

4.3 Write path and persistence

- GUI first: the web UI validates input and restarts the affected services; a raw nvram write does neither.
- Use raw writes only where an item gives them: `nvram set`, `nvram commit`, then the item's service restart. Exception: the I-02 probe never commits.
- Only committed nvram and files on persistent storage (`/jffs` when writable, the SSD) survive a reboot. Runtime changes are wiped: `sysctl`, `/proc` and `/sys` writes, `iptables` rules, `cru` cron jobs, `insmod` modules and runtime `wl` changes; I-02 probes for a boot hook that could reapply them.

5. Setup and snapshot

Snapshots hold secrets (Wi-Fi passphrases, account data): keep them on the host at mode 0600, never on the router's Samba share.

Setup (host, once): add this alias to `~/.ssh/config`; a later port change (L-06) then touches only this entry.

```
Host be92u
    HostName ROUTER_IP
    User USER
    Port SSH_PORT
    IdentityFile ~/.ssh/YOUR_KEY
    IdentitiesOnly yes
```

Save (host; files go to `$XDG_DATA_HOME/be92u`, default `~/.local/share/be92u`):

```
begin
    set -l d ~/.local/share/be92u
    test -n "$XDG_DATA_HOME"; and set d $XDG_DATA_HOME/be92u
    if mkdir -p $d; and chmod 700 $d
        set -l f $d/nvram-(date +%F-%H%M%S).txt
        ssh be92u 'nvram show 2>/dev/null' | sort >$f.part
        set -l ps $pipestatus
        chmod 600 $f.part
        if test $ps[1] -eq 0; and test -s $f.part
            mv $f.part $f; and echo "saved $f"
        else
            echo "snapshot failed (ssh status $ps[1]); kept $f.part" >&2
        end
    else
        echo "cannot prepare $d" >&2
    end
end
```

Compare (host): diff the two newest snapshots.

```
begin
    set -l d ~/.local/share/be92u
    test -n "$XDG_DATA_HOME"; and set d $XDG_DATA_HOME/be92u
    set -l s $d/nvram-*.txt
    if test (count $s) -ge 2
        diff $s[-2] $s[-1]
    else
        echo "need two snapshots" >&2
    end
end
```

Failed runs stay as `.part` files, which the compare step ignores. For a restorable copy, use Administration → Restore/Save/Upload Setting → Save setting [R7] with the remove-password box unticked (a file saved with it ticked must not be imported [R15]), and store it like the snapshots.

6. Collector

Read-only (router): every single-value key the checks read; compare each value with section 2. The web-port keys are context for C-03 and L-02.

```
for k in firmver buildno extendno \
  sshd_enable sshd_port sshd_pass telnetd_enable \
  misc_http_x misc_httpport_x misc_httpsport_x http_enable https_lanport \
  enable_acc_restriction shell_timeout http_autologout ddns_enable_x \
  wps_enable wps_enable_x wan_upnp_enable \
  dnspriv_enable dnspriv_profile dns_priv_override dns_norebind \
  reboot_schedule_enable reboot_schedule_type reboot_schedule; do
  printf '%s=%s\n' "$k" "$(nvram get "$k")"
done
```

7. Checks

7.1 CRIT

C-01 · SSH backdoor indicators (AyySSHush) — CRIT

Risk. AyySSHush, first seen 2025-03-18, exploits the command-injection flaw CVE-2023-39780 to add an attacker SSH key and enable SSH on TCP/53282 for LAN and WAN through ASUS's own settings, so firmware upgrades do not remove it [R1] [R2] [R4]. The attackers also turn off router logging [R4] [R5].

Check (router; syslog path per [R23]):

```
nvram get sshd_port
nvram get sshd_authkeys | tr '>' '\n' | grep -c .
netstat -lnt | grep ':53282'
grep dropbear /tmp/syslog.log*
```

Check (host): fingerprint every authorized key and compare with your own.

```
ssh be92u 'nvram get sshd_authkeys' | tr '>' '\n' | ssh-keygen -lf -
ssh-keygen -lf ~/.ssh/YOUR_KEY.pub
```

Pass when `sshd_port` is the port you chose (never 53282), nothing listens on 53282, no dropbear line shows an unknown address, and the host check prints one fingerprint per stored key, each one yours (`ssh-keygen` skips keys it cannot parse; with none stored, it prints `(stdin) is not a public key file`). No dropbear line at all, not even for your own login, means logging is off; unless you turned it off, count that as an indicator [R4].

Fix (any indicator: treat the router as compromised). Disconnect the WAN cable, then save what the reset destroys: the RAM-only syslog, listeners, cron jobs, `/jffs` and processes (host):

```
set -l f be92u-evidence-(date +%F-%H%M%S).txt
ssh be92u 'cat /tmp/syslog.log*; netstat -lnt; cru l; ls -la /jffs' >$f
ssh be92u 'ps ww 2>/dev/null || ps' >>$f; chmod 600 $f
```

1. Install 102_40497 or newer (H-01): download the image on another machine, verify its SHA-256, upload it manually.
2. Hard factory reset [R8]: Administration → Restore/Save/Upload Setting → Factory default → Restore, with "Restore to factory default settings while clearing all content and data logs" ticked (the default in Merlin's copy [R14]), or the model-specific WPS-button reset [R9]. Either clears all settings plus the AiProtection, Traffic Analyzer and Web History data, which a Restore with the box unticked keeps [R8]; AyySSHush abused AiProtection features [R1].
3. Reconfigure by hand. Never upload a settings file saved after the compromise, including any saved during this audit: the key travels with it (inference from [R1]).
4. Set a new admin password and SSH key pair. Point `IdentityFile` at the new key and clear the router's old host key, which the reset replaced (inference): `ssh-keygen -R ROUTER_IP`, or `ssh-keygen -R '[ROUTER_IP]:SSH_PORT'` for a port other than 22.
5. Rotate every other secret the snapshots hold (Wi-Fi passphrases, WAN login, DDNS, VPN and Samba accounts), then reconnect the WAN.

Verify. From outside the LAN (for example mobile data), TCP/53282 on the WAN address does not answer.

Source conflict. SecurityWeek says the implant survives "factory reboots" [R3]; GreyNoise says firmware upgrades [R1]. A reboot is not a reset: a factory reset clears all settings, the key included [R8].

C-02 · SSH reachable from WAN — CRIT

Risk. `sshd_enable` 1 exposes SSH to the internet and was the AyySSHush end state [R2]. Values: 0 = off, 2 = LAN only, 1 = LAN & WAN [R11].

Check (router): `nvram get sshd_enable`

Pass when 0 or 2.

Fix. GUI: Administration → System → Enable SSH → LAN only.

Verify. From outside the LAN, the SSH port on the WAN address does not answer.

C-03 · Web UI reachable from WAN — CRIT

Risk. `misc_http_x` 1 publishes the web UI on the WAN. On models with HTTPS, this one included, the page allows only HTTPS from WAN, on `misc_httpsport_x`; `misc_httpport_x` serves models without HTTPS [R11]. ASUS's AyySSHush advice for end-of-life models: turn off remote-access features such as SSH, DDNS, AiCloud and Web Access from WAN [R5].

Check (router):

```
nvram get misc_http_x
nvram get misc_httpport_x
nvram get misc_httpsport_x
```

Pass when `misc_http_x` is 0.

Fix. GUI: Administration → System → Enable Web Access from WAN → No.

Verify. From outside the LAN, neither port on the WAN address answers.

7.2 HIGH

H-01 · Firmware currency — HIGH

Risk. 102_39063 fixed an unsafe remote script execution mechanism and a heap buffer overflow; 102_40311 and 102_40497 add more, the latter including one against unintended command execution. 102_38961 brought new defaults (UPnP off, HTTPS on 8443), recommended a factory reset to activate them, and set an admin password policy: at least 10 characters with a letter, a digit and a symbol, no consecutive identical characters [R6]. An in-place upgrade without that reset keeps older values (inference from [R6]); M-05 and L-02 catch them.

Check (router):

```
nvram get firmver
nvram get buildno
nvram get extendno
```

Pass when the version is 3.0.0.6.102_40497 or later: compare `firmver`, then `buildno`, then `extendno` numerically, ignoring any suffix after a hyphen. Last recorded on this router: 102_40311, which fails.

Fix. GUI: Administration → Firmware Upgrade. Before a manual upload, check the download against the SHA-256 that ASUS publishes for 102_40497 [R6] (host):

```
begin
    set -l want dfe0e1b44443a541d80e49c55541cdcd908400acd44b425ad8224f8e9c6a7093
    set -l got (sha256sum FIRMWARE_FILE | string split -f1 -- ' ')
    test "$got" = "$want"; and echo MATCH; or echo MISMATCH
end
```

If the zip does not match, hash the extracted image; ASUS's notes for 102_37435 say to unzip before verifying [R6]. If the admin password predates 102_38961, set one that meets the policy (Administration → System).

H-02 · Telnet daemon — HIGH

Risk. Telnet is an unencrypted root shell on the LAN.

Check (router):

```
nvram get telnetd_enable
netstat -lnt | grep ':23 '
```

Pass when `telnetd_enable` is 0 and nothing listens on port 23.

Fix. GUI: Administration → System → Enable Telnet → No, if the page shows the row; Merlin's copy hides it [R11], and the stock page was not checked. Without the row (router):

```
nvram set telnetd_enable=0
nvram commit
service restart_time
```

The System page applies its settings with `restart_time` [R11], which may drop the SSH session (inference). If port 23 still listens, run `reboot` rather than leave Telnet open until 04:05.

H-03 · WPS — HIGH

Risk. WPS enrolls Wi-Fi clients without the network passphrase.

Check (router):

```
nvram get wps_enable
nvram get wps_enable_x
nvram show 2>/dev/null | grep '_wps_mode='
```

Pass when both keys read 0 and every non-empty `*_wps_mode` reads `disabled`.

Fix. GUI: Wireless → WPS → Enable WPS → off. Raw equivalent, from a script for stock firmware [R19] (router):

```
nvram set wps_enable=0
nvram set wps_enable_x=0
for v in $(nvram show 2>/dev/null | grep '_wps_mode=.' | cut -d= -f1); do
  nvram set "$v=disabled"
done
nvram commit
service restart_wireless
```

`service restart_wireless` drops every Wi-Fi client, an SSH session over Wi-Fi included; run it from a wired client or expect the disconnect (the commit has already run).

7.3 MED

M-01 · SSH password login — MED

Risk. Password login allows guessing from any LAN host.

Check (router):

```
nvram get sshd_pass
nvram get sshd_authkeys | tr '>' '\n' | grep -c .
```

Pass when `sshd_pass` is 0 and the key count is at least 1.

Fix. GUI only: Administration → System. Paste the contents of `YOUR_KEY.pub` into Authorized Keys, apply, and confirm key login with the first Verify command; then set Allow Password Login → No. The page refuses that while Authorized Keys is empty [R11]; a raw nvram write has no such guard and can lock SSH out.

Verify (host):

```
ssh -o PasswordAuthentication=no be92u true; and echo key-login-ok
ssh -o PubkeyAuthentication=no be92u true
```

The first prints `key-login-ok`; the second must fail with `Permission denied` (a password prompt means password login is still on).

M-02 · DoT certificate checks (Strict profile) — MED

Risk. The Opportunistic profile does not authenticate the resolver, so an on-path device can answer in its place. Values: `dnspriv_enable` 1 = DoT; `dnspriv_profile` 1 = Strict, 0 = Opportunistic; each `dnspriv_rulelist` entry is `server>port>hostname>spki` [R12]. In stubby, Strict is `tls_authentication: GETDNS_AUTHENTICATION_REQUIRED` with TLS as the only transport [R24].

Check (router; stubby path per [R22]):

```
nvram get dnspriv_enable
nvram get dnspriv_profile
nvram get dnspriv_rulelist | tr '<' '\n'
nvram get dnspriv_rulelist | tr '<' '\n' |
  awk -F'>' 'NF && $3 == "" && $4 == "" {print "NO HOSTNAME OR PIN:", $1}'
grep -E '^[^#]*(tls_auth|GETDNS_TRANSPORT)' /etc/stubby/stubby.yml
```

Pass when `dnspriv_enable` and `dnspriv_profile` are 1, no `NO HOSTNAME OR PIN` line prints, and `stubby.yml` shows `GETDNS_AUTHENTICATION_REQUIRED` with `GETDNS_TRANSPORT_TLS` as its only transport.

> **Precondition for Strict.** Each upstream needs a TLS Hostname (stubby's `tls_auth_name`) or an SPKI Fingerprint so it can be authenticated [R24]; one without fails authentication. The `NO HOSTNAME OR PIN` check must print nothing before the switch. AdGuard's default servers (`94.140.14.14`, `94.140.15.15`, `2a10:50c0::ad1:ff`, `2a10:50c0::ad2:ff`) use the TLS Hostname `dns.adguard-dns.com` [R25]; `dns.adguard.com` is the 2018 name [R26].

Fix. GUI: WAN → Internet Connection → DNS Privacy Protocol → DNS-over-TLS (DoT), then DNS-over-TLS Profile → Strict. Raw equivalent of the profile switch alone, with DoT already on and its server list filled (router; the GUI adds `restart_stubby` [R12]):

```
nvram set dnspriv_profile=1
nvram commit
service restart_stubby
```

Verify (host): a fresh random name, which no cache holds, returns `rcode: NXDOMAIN`; `SERVFAIL` means no upstream answered under Strict.

```
drill (random)-(date +%s).example.com @ROUTER_IP | grep rcode
```

Rollback (router), on `SERVFAIL` or if DNS stops resolving:

```
nvram set dnspriv_profile=0
nvram commit
service restart_stubby
```

Trade-off. Strict drops an upstream that fails authentication instead of using it unauthenticated. The host has no other resolver by design, so if every upstream fails authentication, the LAN has no DNS until the rollback. An AdGuard outage stops DNS under either profile while TLS is the only transport [R24].

M-03 · DNS rebind protection — MED

Risk. Without it, a public name can resolve to a LAN address and expose LAN services, the web UI included, to hostile web pages. `dns_norebind` 1 = on [R12].

Check (router): `nvram get dns_norebind`

Pass when 1.

Fix. GUI: WAN → Internet Connection → Enable DNS Rebind protection → Yes. Raw: `nvram set dns_norebind=1`, `nvram commit`, then `service restart_dnsmasq` [R12]. Public names that point at private addresses then stop resolving.

M-04 · Samba guest access — MED

Risk. With guest login on, anyone on the network reaches the share without a password [R10]. Last recorded on this router: guest login on.

Check (router; `smb.conf` path per [R22]):

```
grep -inE '^\[|guest|protocol' /etc/smb.conf
```

Check (host): an anonymous listing (`-U%`: empty user and password) of each share the router check lists (not `[global]`).

```
smbclient -U% //ROUTER_IP/SHARE -c ls
```

Pass when no share has `guest ok = yes` and every anonymous listing fails with `NT_STATUS_ACCESS_DENIED` or `NT_STATUS_LOGON_FAILURE`; a protocol-negotiation error proves nothing. SMBv1 is not graded (section 9).

Fix. GUI: USB Application → Servers Center → Network Place (Samba) Share / Cloud Disk → Allow guest login → off [R10]. Clients that used guest access need an account afterwards.

M-05 · UPnP — MED

Risk. UPnP lets any LAN program open inbound ports. `wan_upnp_enable` 1 = on [R12]; ASUS ships it off from 102_38961 [R6].

Check (router): `nvram get wan_upnp_enable`

Pass when 0. An application that needs it is an ACCEPTED exception; name it under this item.

Fix. GUI: WAN → Internet Connection → Enable UPnP → No. For a game that needs inbound ports, prefer explicit port forwards.

7.4 LOW

L-01 · Browser auto-DoH bypass — LOW

Risk. Browsers that turn on DoH by themselves bypass the router and its AdGuard upstream. Prevent client auto DoH (`dns_priv_override`): 0 = Auto, 1 = Yes, 2 = No [R12]. Auto, the default, applies the block only while DNS Privacy or DNS Filter in global mode is on [R15]; Yes applies it always (inference).

Check (router): `nvram get dns_priv_override`

Pass when 1, or 0 while `dnspriv_enable` is 1 (Auto then blocks too).

Fix. GUI: WAN → Internet Connection → Prevent client auto DoH → Yes. Raw: `nvram set dns_priv_override=1`, `nvram commit`, then `service restart_dnsmasq` [R12]. DoH configured by hand in a browser still bypasses the router (inference).

L-02 · Web UI over plain HTTP — LOW

Risk. HTTP sends the admin session in clear text on the LAN. `http_enable` 0 = HTTP, 1 = HTTPS, 2 = both [R11]; HTTPS on 8443 is the default from 102_38961 [R6].

Check (router):

```
nvram get http_enable
nvram get https_lanport
```

Pass when `http_enable` is 1.

Fix. GUI: Administration → System → Authentication Method → HTTPS.

Verify. The web UI loads over HTTPS on the `https_lanport` port.

L-03 · Admin access allowlist — LOW

Risk. Any LAN device can reach the web UI and SSH. With `enable_acc_restriction` 1, only addresses in `restrict_rulelist` get in; each entry is `enabled>address>type`, and type is a bitmask: 1 = Web UI, 2 = SSH, 4 = Telnet [R11].

Check (router):

```
nvram get enable_acc_restriction
nvram get restrict_rulelist | tr '<' '\n'
```

Pass when `enable_acc_restriction` is 1 and the host's address has an enabled entry of type 3 (Web UI and SSH).

Fix. First give the host a DHCP reservation so its address cannot drift and lock you out. Then GUI: Administration → System → Enable Access Restrictions → Yes; add the host with Web UI and SSH, then every other device that manages the router with Web UI. The list holds at most four entries and needs at least one enabled Web UI entry [R11]; anything not listed is refused.

Verify. Another LAN device is refused.

L-04 · Idle timeouts — LOW

Risk. Sessions left open stay usable. `shell_timeout` is in seconds and `http_autologout` in minutes; 0 disables either [R11].

Check (router):

```
nvram get shell_timeout
nvram get http_autologout
```

Pass when both are non-zero.

Fix. GUI: Administration → System → Idle Timeout (SSH) and Auto Logout (web UI). Both fields take minutes: 10 to 999, or 0 for off [R11].

L-05 · DDNS exposure — LOW

Risk. DDNS publishes a stable name for the WAN address; ASUS lists it among the remote-access features to turn off (C-03) [R5].

Check (router): `nvram get ddns_enable_x`

Pass when 0. Active use is an ACCEPTED exception; note what depends on it.

Fix. GUI: WAN → DDNS, switch DDNS off. Anything built on it (remote app access, DDNS-based certificates) stops working (inference).

L-06 · SSH default port — LOW

Risk. Port 22 draws scans; ASUS's UI strings suggest a port from 1024 to 65535 [R15]. Impact is low once C-02 passes.

Check (router): `nvram get sshd_port`

Pass when 1024 to 65535 and not 53282 (C-01).

Fix. GUI: Administration → System → SSH Port. The open session drops when dropbear restarts (inference); then update `Port` in the `be92u` host alias (section 5).

Verify. `ssh be92u true` succeeds on the new port.

L-07 · USB SSD TRIM — LOW

Risk. Without TRIM, the Lexar ES3 USB SSD's write performance and wear leveling degrade over time.

> **Precondition.** A current backup of the SSD: the check's `fstrim` sends UNMAP, and a USB bridge that mishandles it can corrupt data (inference).

Check (router; adjust `sda` and `/tmp/mnt/sda1` to what `mount | grep /tmp/mnt` shows):

```
cat /sys/block/sda/queue/discard_max_hw_bytes
cat /sys/block/sda/device/scsi_disk/*/provisioning_mode
type fstrim && fstrim -v /tmp/mnt/sda1
```

Pass when `discard_max_hw_bytes` is above 0 and `fstrim -v` exits without an error (0 bytes trimmed is valid right after a trim). Record ACCEPTED, with the reason, if `type fstrim` fails (BusyBox without the applet [R20]), if TRIM fails even after the fix, or if it works only after the fix, which a reboot or re-plug reverts.

Fix. Some USB storage passes TRIM only after `provisioning_mode` is set to `unmap` [R20]; the write is RAM-only and reverts at the next reboot or re-plug, such as the one in I-02 (router):

```
for f in /sys/block/sda/device/scsi_disk/*/provisioning_mode; do
  echo unmap > "$f"
done
fstrim -v /tmp/mnt/sda1
```

7.5 INFO

I-01 · Persistence, reboot schedule — INFO

`reboot_schedule` holds seven day flags, Sunday first, then HHMM; a daily 04:05 reboot reads `11111110405`. With `reboot_schedule_type` 1, newer builds keep a day-of-month schedule in `reboot_schedule_month` [R11].

Check (router):

```
nvram get reboot_schedule_enable
nvram get reboot_schedule_type
nvram get reboot_schedule
mount | grep jffs
```

Expected. `reboot_schedule_enable` 1, `reboot_schedule_type` 0 or empty (`nvram get` prints nothing for an unset key [R16]), `reboot_schedule` `11111110405`. Record whether `/jffs` mounts `ro` or `rw` (last recorded: read-only); I-02 depends on it.

I-02 · Boot-hook probe — INFO

Stock firmware has no supported startup hook: newer official firmware blocks the `script_usbmount` hook [R17], and asd, ASUS's security daemon, most likely clears the variable [R18].

Probe (router; procedure from [R17]; the probe never commits, so a reboot also clears it):

```
nvram set script_usbmount="/bin/touch /tmp/yesitworks"
sleep 15; nvram get script_usbmount
```

Empty output means blocked; skip to the reset. Otherwise eject the SSD in the GUI (the Samba share goes offline until it is back), plug it back in, then:

```
cat /tmp/yesitworks
```

A missing file also means blocked.

Reset (router; run in either case):

```
nvram set script_usbmount=
```

Fallback, unverified on this model: the asusware workaround needs Download Master and an `asusware.arm` folder at the drive root, whose init script runs `/jffs/scripts/usb-mount-script` [R18]; with a read-only `/jffs` (I-01) it would have to call a file on the SSD (inference).

I-03 · Thermals, radios, clients — INFO

Collect (router):

```
awk '{print "cpu", $1/1000, "C"}' /sys/class/thermal/thermal_zone0/temp
for p in /sys/class/net/*; do
  i=${p##*/}
  case $i in *.*) continue ;; esac
  wl -i "$i" status 2>/dev/null | grep -q GHz || continue
  echo "== $i"
  wl -i "$i" status | grep GHz
  wl -i "$i" phy_tempsense | awk '{print $1/2+20, "C approx"}'
  wl -i "$i" assoclist
done
```

Per-client detail, for example for the 6 GHz host: `wl -i IFACE sta_info CLIENT_MAC`.

The thermal path, band match and /2+20 conversion come from a stock-firmware script [R21] in a collection mostly tested on an RT-AX58U v2 [R17]. The loop skips virtual (guest) interfaces.

I-04 · Listening services baseline — INFO

Collect (router):

```
netstat -lntup 2>/dev/null || netstat -lntu
```

Save the output; on later audits, any new listener needs an explanation.

8. Do not apply

- Regional, transmit-power and ignore-DFS nvram overrides from user-posted scripts, one aimed at this model [R27]: they bypass regulatory limits and can destabilize the radios (inference).
- `stop_fsck=1`: its author warns against it on a disk used for file sharing [R18].
- `nvram unset`, `nvram erase` and raw writes to the flash partitions: no item needs them.

9. Limitations and open items

- nvram semantics and GUI labels come from ASUS's web-UI code and strings as carried in Asuswrt-Merlin's tree [R11] [R12] [R14] [R15]; stock firmware was not diffed. After any write, confirm that the GUI shows the new value.
- Not verified on this model: the config and syslog paths taken from stock-firmware scripts [R22] [R23], radio interface names and band match, the thermal path and /2+20 conversion, Download Master availability, the asusware architecture suffix, the `fstrim` applet and the Lexar ES3's UNMAP support.
- SMBv1 is not graded: Merlin's Samba page sets it with `smbd_protocol` (0 = SMBv1, 1 = SMBv2, 2 = both) [R13]; the stock page and its `smb.conf` output were not verified. The M-04 grep shows any protocol lines.
- AiCloud, which ASUS lists with the remote-access features [R5], is not checked; if the GUI offers it, keep it off.

- ASUS's nvram tool lists `save FILE` and `restore FILE` [R16]; untested here, so restorable copies come from the GUI's Save setting (section 5).

10. References

10.1 AyySSHush reporting		
R1	GreyNoise: AyySSHush analysis	https://www.labs.greynoise.io/grimoire/2025-03-28-ayysshush/
R2	SecurityAffairs: AyySSHush payload	https://securityaffairs.com/178413/malware/new-ayysshush-botnet-compromised-over-9000-asus-routers-adding-a-persistent-ssh-backdoor.html
R3	SecurityWeek: 9,000 routers backdoored	https://www.securityweek.com/greynoise-flags-9000-asus-routers-backdoored-via-patched-vulnerability/
R4	BleepingComputer: AyySSHush, CVE-2023-39780	https://bleepingcomputer.com/news/security/botnet-hacks-9-000-plus-asus-routers-to-add-persistent-ssh-backdoor
R5	Guru3D: NVRAM backdoors, ASUS advice	https://www.guru3d.com/story/asus-routers-vulnerable-hackers-implant-undetectable-backdoors-in-nvram/

10.2 ASUS documentation		
R6	ASUS: RT-BE92U firmware and release notes	https://www.asus.com/supportonly/rt-be92u/helpdesk_bios/
R7	ASUS FAQ: back up router settings	https://www.asus.com/support/faq/1001376/
R8	ASUS FAQ: factory default reset	https://www.asus.com/support/FAQ/1000925
R9	ASUS FAQ: hard reset, method 1 (WPS)	https://www.asus.com/support/FAQ/1039077/
R10	ASUS FAQ: Network Place (Samba) Share	https://www.asus.com/support/FAQ/1011279/

10.3 ASUS web UI and tools (Asuswrt-Merlin tree)		
R11	System page (Advanced_System_Content.asp)	https://github.com/RMerl/asuswrt-merlin.ng/blob/master/release/src/router/www/Advanced_System_Content.asp
R12	WAN page (Advanced_WAN_Content.asp)	https://github.com/RMerl/asuswrt-merlin.ng/blob/master/release/src/router/www/Advanced_WAN_Content.asp
R13	Samba page (Advanced_AiDisk_samba.asp)	https://github.com/RMerl/asuswrt-merlin.ng/blob/master/release/src/router/www/Advanced_AiDisk_samba.asp
R14	Restore page (Advanced_SettingBackup_Content.asp)	https://github.com/RMerl/asuswrt-merlin.ng/blob/master/release/src/router/www/Advanced_SettingBackup_Content.asp
R15	English UI strings (EN.dict)	https://github.com/RMerl/asuswrt-merlin.ng/blob/master/release/src/router/www/EN.dict
R16	nvram tool (nvram/nvram.c)	https://github.com/RMerl/asuswrt-merlin.ng/blob/master/release/src/router/nvram/nvram.c

10.4 Scripts for stock firmware (jacklul/asuswrt-scripts)		
R17	README: script_usbmount probe, tested models	https://github.com/jacklul/asuswrt-scripts
R18	asusware-usb-mount-script README	https://github.com/jacklul/asuswrt-scripts/tree/master/asusware-usb-mount-script
R19	disable-wps.sh	https://github.com/jacklul/asuswrt-scripts/blob/master/scripts/disable-wps.sh
R20	fstrim.sh	https://github.com/jacklul/asuswrt-scripts/blob/master/scripts/fstrim.sh
R21	temperature-warning.sh	https://github.com/jacklul/asuswrt-scripts/blob/master/scripts/temperature-warning.sh
R22	custom-configs.sh: stock config paths	https://github.com/jacklul/asuswrt-scripts/blob/master/scripts/custom-configs.sh
R23	service-event.sh: syslog path	https://github.com/jacklul/asuswrt-scripts/blob/master/scripts/service-event.sh

10.5 DNS		
R24	stubby: stubby.yml.example (Strict)	https://github.com/getdnsapi/stubby/blob/develop/stubby.yml.example
R25	AdGuard DNS: public DNS servers	https://adguard-dns.io/en/public-dns.html
R26	AdGuard: DNS launch announcement (2018)	https://adguard-dns.io/en/blog/adguard-dns-announcement.html

10.6 Other		
R27	Gist: regional, TX-power, DFS overrides	https://gist.github.com/francoism90/3dede7973354d067c41bff5e54203fe9